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INDUSTRIAL VEHICLE

Abstract

An industrial vehicle capable of manned travel and unmanned travel by the automated operation includes a vehicle body; and an automated operation unit that is mounted on the vehicle body and in which a plurality of devices for automated operation is integrated. The plurality of devices includes: an automated operation controller controlling the automated operation; a display device on which information related to the automated operation is displayed; an input device on which data related to the automated operation is input; and an emergency stop device causing an emergency stop of the automated operation. The automated operation unit includes an accommodation case accommodating the automated operation controller, the display device, the input device, and the emergency stop device. The accommodation case is positioned in the vehicle body at a height equal to or lower than a height that does not obstruct an operator's view during the manned travel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031490 filed on Mar. 1, 2024, the entire disclosure of which is incorporated herein by reference.

[0002] The present disclosure relates to an industrial vehicle, and more particularly to an industrial vehicle capable of unmanned and manned travel.

BACKGROUND ART

[0003] In recent years, unmanned travel by automated operation of an industrial vehicle such as a forklift truck has been realized, and Japanese Patent Publication No. 5856362 discloses a forklift truck for manned and unmanned operation. For this type of the forklift truck, an existing man-operated forklift truck is modified to achieve the automated operation. Therefore, devices necessary for the automated operation are additionally mounted on the existing man-operated forklift truck.

[0004] However, since there is not enough space in a vehicle body of the man-operated forklift truck, it is not possible to accommodate all devices (except for sensors) necessary for automated operation inside the forklift truck for the manned and unmanned operation. For this reason, it is necessary to consider where and how to install the devices necessary for the automated operation. For example, in a case of a forklift truck, it is possible to install the equipment necessary for the automated operation in a head guard that protects the driver's seat, but there is a risk that the devices necessary for the automated operation may obstruct the operator's view when the vehicle is being operated by an operator. In addition, since the devices are installed at high positions, it is difficult to perform maintenance work on the devices. Although it is possible to provide the devices necessary for the automated operation inside the counterweight, it requires a dedicated counterweight, which increases manufacturing costs.

[0005] The present disclosure has been made in view of the above problems, and is directed to providing an industrial vehicle that secures an operator's view during manned operation and has excellent workability on an automated operation unit.

SUMMARY

[0006] In accordance with an aspect of the present disclosure, there is provided an industrial vehicle capable of manned travel and unmanned travel by the automated operation. The industrial vehicle includes a vehicle body; and an automated operation unit that is mounted on the vehicle body and in which a plurality of devices for automated operation is integrated. The plurality of devices includes an automated operation controller that controls the automated operation, a display device on which information related to the automated operation is displayed, an input device on which data related to the automated operation is input, and an emergency stop device that causes an emergency stop of the automated operation. The automated operation unit includes an accommodation case that accommodates the automated operation controller, the display device, the input device, and the emergency stop device. The accommodation case is positioned in the vehicle body at a height equal to or lower than a height that does not obstruct an operator's view during the manned travel.

[0007] Other aspects and advantages of the disclosure will become apparent from the following

description, taken in conjunction with the accompanying drawings, illustrating by way of example the principles of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The disclosure, together with objects and advantages thereof, may best be understood by reference to the following description of the embodiments together with the accompanying drawings in which:

[0009] FIG. 1 is a side view of a forklift truck according to a first embodiment;

[0010] FIG. 2 is a perspective view of the forklift truck according to the first embodiment;

[0011] FIG. 3 is a perspective view of a main part of the forklift truck according to the first embodiment, as viewed from the rear left;

[0012] FIG. 4 is a perspective view of the main part of the forklift truck according to the first embodiment, as viewed from the rear right;

[0013] FIG. 5 is a perspective view illustrating devices accommodated in an automated operation unit;

[0014] FIG. 6 is a perspective view illustrating a state in which the automated operation unit is displaced rearward relative to a right pillar; and

[0015] FIG. 7 is a side view of a forklift truck according to a second embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

First Embodiment

[0016] Hereinafter, an industrial vehicle according to a first embodiment will be described with reference to the accompanying drawings. The industrial vehicle of the present embodiment is a battery-powered forklift truck that is capable of unmanned travel by an automated operation, and manned travel driven by an operator. In the following description, “front-rear”, “left-right”, and “up-down” directions are defined with respect to an operator sitting in a driver's seat of the forklift truck and facing the forward direction of the forklift truck. Firstly, the forklift truck will be described.

[0017] As illustrated in FIGS. 1 and 2, a vehicle body **11** of a forklift truck **10** includes a main body **12**. The forklift truck **10** includes front wheels **13** provided in a front of the main body **12**, and rear wheels **14** provided in a rear of the main body **12**. The front wheels **13** are driving wheels and the rear wheels **14** are steered wheels. A cargo handling device **15** is disposed in the front of the main body **12**, and a counterweight **16** is disposed in the rear of the main body **12**. A driver's seat **17** is provided near the center of the main body **12**. The main body **12** is equipped with a travel motor **18** as a driving power source for traveling, and also a battery **19** for storing electric power for driving the travel motor **18**.

[0018] The cargo handling device **15** includes a mast **22** having outer masts **20** and inner masts **21**. A pair of left and right outer masts **20** is provided with the inner masts **21** which are slidable inside the outer masts **20**. The mast **22** includes a lift cylinder **23** of a single-action type that is operated by hydraulic oil. With the operation of the lift cylinder **23**, the inner masts **21** slide to move up and down within the outer masts **20**.

[0019] A pair of left and right forks **24** for supporting a cargo **W** is attached to the mast **22** via a lift bracket **25**. The lift bracket **25** is configured to move up and down relative to the inner mast **21**. A tilt cylinder **26** of a reciprocating type that is operated by hydraulic oil is provided between the main body **12** and the outer masts **20**. The mast **22** tilts forward and backward with the lower end of the mast **22** as a fulcrum with an operation of the tilt cylinder **26**. The tilt cylinder **26** is operated by supplying and discharging hydraulic oil.

[0020] The counterweight **16** is disposed in the rear of the main body **12**. The counterweight **16** is

used for adjustment of the weight of the forklift truck **10** and weight balance of the vehicle body **11**. As illustrated in FIGS. **3** and **4**, the counterweight **16** has a rear surface **27** that is curved in an arc shape so as to protrude rearward in a plan view, a pair of left and right side surfaces **28**, and a flat top surface **29**. As illustrated in FIGS. **1** and **2**, a wheel house **30** for the rear wheels **14** is disposed across the main body **12** and the counterweight **16**.

[0021] As illustrated in FIGS. **1** and **2**, a head guard **31** is disposed above the driver's seat **17**. The head guard **31** is supported by a pair of left and right front pillars **32** disposed in the front of the main body **12** and a pair of left and right rear pillars **33** provided in the rear of the main body **12**. The main body **12** has a toe board **34** that forms the floor surface of the driver's seat **17**. The battery **19** is disposed below the toe board **34**. As illustrated in FIG. **2**, an instrument panel **35** is provided in front of the toe board **34**. The instrument panel **35** has a steering column **36** and a plurality of cargo handling levers **37**. The steering column **36** supports a steering wheel **38**.

[0022] A seat stand **39** is disposed behind the toe board **34** on the main body **12**. The seat stand **39** is openable and closable. A driver's seat **40** is mounted on the seat stand **39**. The seat stand **39** has a recess **41** that is recessed rearward (see FIG. **1**) in the front portion of the seat stand **39**. When the battery **19** is to be replaced, the toe board **34** is removed and the seat stand **39** is rotated toward the counterweight **16**, which allows the battery **19** to be removed from the main body **12**. The vehicle body **11** of the present embodiment includes at least the counterweight **16**, the head guard **31**, the front pillars **32**, the rear pillars **33**, the toe board **34**, the instrument panel **35**, and the seat stand **39**, in addition to the main body **12**.

[0023] The forklift truck **10** of the present embodiment includes an automated operation unit **42** mounted on the counterweight **16**, which is a part of the vehicle body **11**, and behind the driver's seat **40**. In the present embodiment, the automated operation unit **42** is added to a forklift truck of an existing design that can be driven by an operator to realize the forklift truck **10** capable of manned and unmanned operation. In order to achieve the automated operation of the forklift truck **10**, a plurality of devices for automated operation that enables unmanned travel (hereinafter referred to as a device group for automated operation) is required. Most of the device group for the automated operation is integrated in the automated operation unit **42**.

[0024] As illustrated in FIGS. **3**, **4**, and **5**, the automated operation unit **42** includes an accommodation case **43** that accommodates the device group for the automated operation. The accommodation case **43** has a bottom **44**, a front plate **45**, an upper plate **46**, a right plate **47**, a left plate **48**, and a rear plate **49**. The bottom **44** is the bottom of the accommodation case **43** and is formed by a frame. As illustrated in FIG. **6**, the front plate **45** is formed of a plate member extending vertically from a front portion of the bottom **44**. The upper plate **46** is formed of a plate member that is connected to an upper portion of the front plate **45**, and forms an upper surface of the accommodation case **43**. The upper plate **46** is inclined downward from the front to the rear. Specifically, the upper plate **46** is inclined so that the automated operation unit **42** is positioned at a height equal to or lower than a predetermined height, and equal to or lower than a height of the driver's seat **40**. The height and the inclination of the upper plate **46** are set so that the automated operation unit **42** does not obstruct a rear view of the operator during the manned operation. Thus, the accommodation case is positioned in the vehicle body at a height equal to or lower than a height that does not obstruct an operator's view during the manned travel. The front plate **45** and the upper plate **46** are integrally formed.

[0025] As illustrated in FIG. **4**, the right plate **47** is formed of a plate member located on the right side of the accommodation case **43**. As illustrated in FIGS. **3** and **6**, the left plate **48** is formed of a plate member located on the left side of the accommodation case **43**. The rear plate **49** is formed of a plate member located at the rear of the accommodation case **43**. The rear plate **49** has a surface that is curved in an arc shape so as to protrude rearward in a plan view, similarly to the rear surface **27** of the counterweight **16**. In addition, the accommodation case **43** is disposed on the counterweight **16** so as to be located inside an outer edge of the counterweight **16** in a plan view.

The rear plate **49**, the right plate **47**, and the left plate **48** are integrally formed, and are detachable from the bottom **44**, the front plate **45**, and the upper plate **46**.

[0026] As illustrated in FIGS. **3** and **4**, a base **50** that supports the accommodation case **43** is provided on the counterweight **16**. As illustrated in FIG. **4**, a rotary shaft **51** is provided in the base **50** near the right rear pillar **33**. The axis of the rotary shaft **51** extends in the up-down direction, and a right end of the bottom **44** is attached to the rotary shaft **51**. Therefore, the accommodation case **43** is rotatable generally horizontally about the rotary shaft **51** and is displaceable relative to the vehicle body **11**. As illustrated in FIG. **6**, when the accommodation case **43** is in a maximum displacement state, the accommodation case **43** is located behind the right rear pillar **33**. The base **50** and the rotary shaft **51** correspond to a displacement mechanism that allows the accommodation case **43** to be displaced relative to the vehicle body **11**. The base **50** has a stopper **52** that restricts rotary motion of the accommodation case **43** near the left rear pillar **33** (see FIGS. **3** and **6**). The accommodation case **43** has a protrusion **53** at a left end of the bottom **44**, and the protrusion **53** serves as an engaged portion to be engaged with the stopper **52**. Disengaging the stopper **52** allows the accommodation case **43** to rotate.

[0027] Next, the device group for the automated operation integrated in the accommodation case **43** will be described. As illustrated in FIG. **5**, the device group for the automated operation includes an automated operation controller **55**, a touch panel **56**, a changeover switch device **57**, an emergency stop device **58**, a PLC unit **59**, a DC-DC converter **60**, a wireless communication device **61**, a multilayer indication light **62**, an emergency stop receiver **63**, and a hub **64**. The automated operation controller **55** is a device that controls devices related to the automated operation, and includes a CPU (not illustrated) executing various calculations and control programs, and a memory (not illustrated) storing various data and programs. Thus, the automated operation controller **55** controls the automated operation. The automated operation controller **55** is fixed to the bottom **44** of the accommodation case **43**. The automated operation controller **55** is connected to the touch panel **56**, the changeover switch device **57**, the emergency stop device **58**, the PLC unit **59**, the DC-DC converter **60**, the wireless communication device **61**, the multilayer indication light **62**, the emergency stop receiver **63**, and the hub **64**. In addition, the automated operation controller **55** is connected to an on-board controller (not illustrated) mounted on the main body **12**. The on-board controller has a function of controlling the travel motor **18** and the cargo handling device **15** during both manned and unmanned travel.

[0028] The touch panel **56** has a panel screen **65** that is capable of displaying various types of information and on which information can be input with a touch (see FIGS. **4** and **5**). Thus, the touch panel **56** corresponds to an input device on which data necessary for the automated operation can be input, and corresponds to a display device that displays information related to the automated operation. In the present embodiment, the touch panel **56** is attached to the accommodation case **43** so that the panel screen **65** is exposed at the right plate **47**. By inputting information on the panel screen **65** of the touch panel **56** by the operator, for example, an emergency stop state may be released.

[0029] The changeover switch device **57** is a device equipped with a switch **66** for switching and selecting the automated operation or manual operation. In the present embodiment, the changeover switch device **57** is attached to the accommodation case **43** so that the switch **66** is exposed at the right plate **47**. The emergency stop device **58** is a device for causing emergency stop of the unmanned travel of the forklift truck **10**, and is provided with emergency stop buttons **67**. In the present embodiment, the emergency stop buttons **67** are provided so as to protrude from the rear plate **49**, the right plate **47**, and the left plate **48**, respectively, of the accommodation case **43**. If any of the emergency stop buttons **67** is pressed during the unmanned travel by the automated operation, the forklift truck **10** immediately stops.

[0030] The PLC unit **59** is a device in which a plurality of PLCs (programmable logic controllers) are integrated to form a unit. The DC-DC converter **60** is a device for changing a DC voltage and

performing voltage conversion. In the present embodiment, the DC-DC converter **60** is attached to the front plate **45** of the accommodation case **43**. The wireless communication device **61** is provided for wireless communication with a host system (not illustrated) that manages the operation of the forklift truck **10** and the like.

[0031] The multilayer indication light **62** is a rod-shaped indicator light equipped with a plurality of lights arranged in the up-down direction, and has a red light **68**, a yellow light **69**, and a green light **70** arranged in this order from the top. The red light **68** is turned on when the forklift truck **10** is brought to an emergency stop. The yellow light **69** is turned on when an abnormality occurs in the forklift truck **10**, and the green light **70** is turned on when the forklift truck **10** operates normally. In this embodiment, the multilayer indication light **62** is provided upright on the upper plate **46** of the accommodation case **43** near the right rear pillar **33**.

[0032] The emergency stop receiver **63** is a receiver that receives a signal transmitted from the emergency stop device **58** when any of the emergency stop buttons **67** is pressed. The hub **64** is a device in which cables for communication between the automated operation controller **55** and devices are integrated. In the present embodiment, the PLC unit **59**, the DC-DC converter **60**, the wireless communication device **61**, the emergency stop receiver **63**, and the hub **64** are attached to the front plate **45** of the accommodation case **43**. It is noted that power wiring and communication control wiring are pulled out together from near the right rear pillar **33** of the accommodation case **43**.

[0033] The device group for the automatic operation includes a plurality of devices that are not integrated in the automated operation unit **42**. Examples of the plurality of devices that are not integrated into the automated operation unit **42** include sensors provided in various parts of the vehicle body **11**. In the present embodiment, such sensors include a plurality of obstacle detection sensors **71** protruding outward from the rear surface **27** of the counterweight **16**, in addition to a front camera (not illustrated) and an obstacle detection sensor (not illustrated) provided in the cargo handling device **15** (see FIG. 2). Sensors of this type are connected to the automated operation controller **55**. The obstacle detection sensors **71** are disposed so as to face rearward.

[0034] Next, the operation of the forklift truck **10** of the present embodiment will be described. When the forklift truck **10** is to perform unmanned travel by the automated operation, firstly, the operator sets the switch **66** of the changeover switch device **57** in the automated operation unit **42** to an automated operation position. Then, the operator performs necessary operations on the touch panel **56** to start the automated operation. Thus, the forklift truck **10** starts the automated operation and travels along a preset travel route. The forklift truck **10** travels while detecting an obstacle around the vehicle body **11**.

[0035] During the unmanned travel, when the forklift truck **10** operates normally, only the green light **70** of the multilayer indication light **62** is kept turned on. When any abnormality occurs in the forklift truck **10**, only yellow light **69** of the multilayer indication light **62** is turned on. When the forklift truck **10** is at the emergency stop, only red light **68** is turned on. It is noted that the forklift truck **10** is brought to the emergency stop if the any of the emergency stop buttons **67** is pressed during the unmanned travel.

[0036] When the forklift truck **10** is operated by an operator, the operator sets the switch **66** of the changeover switch device **57** in the automated operation unit **42** to a manual operation position. Then, the operator is seated on the driver's seat **40** and performs manned operation. When the forklift truck **10** moves backward during the manned operation, the operator checks the rear. The automated operation unit **42** is positioned at a height equal to or lower than the height of the driver's seat **40**, and the automated operation unit **42** hardly obstructs the operator's rear view. In addition, since the automated operation unit **42** is mounted on the counterweight **16** so as to be positioned inside the outer edge of the counterweight **16** in a plan view, it is less likely to interfere with obstacles around the vehicle body **11** when the forklift truck **10** is travelling backward or turning.

[0037] Since the forklift truck **10** has the battery **19**, when charging or replacing the battery, a space above the battery **19** has to be opened by removing the toe board **34** and rotating the seat stand **39** rearward to stand it up. In rotating the seat stand **39**, the operator releases the stopper **52** so that the automated operation unit **42** mounted on the counterweight **16** does not obstruct rotating of the seat stand **39**, and the automated operation unit **42** is rotated to be positioned behind the right rear pillar **33** (see FIG. 6). When charging or replacement of the battery **19** is completed, the operator folds down the seat stand **39**. After the seat stand **39** is folded down, the automated operation unit **42** is rotated so as to be positioned on the counterweight **16**, and the protrusion **53** is engaged with the stopper **52** to fix the automated operation unit **42**.

[0038] Maintenance work will be performed on the various devices integrated in the automated operation unit **42**. However, since the automated operation unit **42** is installed so as to be positioned at a height equal to or lower than the height of the driver's seat **40**, a maintenance worker can easily perform maintenance without having to work at height using a stepladder or the like.

[0039] The forklift truck **10** of the present embodiment offers the following advantageous effects.

[0040] (1) The accommodation case **43** that accommodates the automated operation controller **55**, the touch panel **56**, and the emergency stop device **58** is mounted on the vehicle body **11** so as to be positioned at a height equal to or lower than the predetermined height. This configuration prevents the automated operation unit **42** from obstructing the operator's view, and secures the operator's view during the manned operation. In addition, positioning the automated operation unit **42** at the height equal to or lower than the predetermined height allows maintenance for the automated operation unit **42** to be performed easily, which increases the workability for the automated operation unit **42**. [0041] (2) The vehicle body **11** has the main body **12**, the driver's seat **40** provided on the main body **12**, and the counterweight **16** provided in the rear of the main body **12**, and the accommodation case **43** is mounted on the counterweight **16** and behind the driver's seat **40**. Therefore, when the forklift truck **10** moves forward during the manned operation, the automated operation unit **42** does not obstruct the operator's forward view. In addition, even when the forklift truck **10** moves backward during the manned operation, the automated operation unit **42** is prevented from obstructing the operator's rear view. [0042] (3) The accommodation case **43** is positioned at a height equal to or lower than the height of the driver's seat **40**, which prevents the automated operation unit **42** from obstructing the rear view of the operator even when the forklift truck moves backward during the manned operation. [0043] (4) The accommodation case **43** is located inside the outer edge of the counterweight **16** in a plan view, which suppresses interference between the automated operation unit **42** and obstacles around the vehicle body **11** while the forklift truck **10** travels. [0044] (5) The accommodation case **43** has the base **50** and the rotary shaft **51** functioning as the displacement mechanism that allows the accommodation case **43** to be displaced relative to the vehicle body **11**. Thus, the accommodation case **43** can be displaced relative to the vehicle body **11** by the displacement mechanism. As a result, even when the driver's seat **40** is to be tilted backward, the automated operation unit **42** interfering with the backward tilting of the driver's seat **40** may be avoided. [0045] (6) Since the emergency stop device **58** has the emergency stop buttons **67** protruding from the surface of the accommodation case **43**, the emergency stop buttons **67** do not have to be provided in the vehicle body **11**. As a result, there is no need to process the vehicle body **11** or run wiring to install the emergency stop buttons **67**. In addition, since the emergency stop buttons **67** are provided on the accommodation case **43**, there are almost no restrictions on the positions of the emergency stop buttons **67**, and the positions of the emergency stop buttons **67** are intuitively easy to identify. [0046] (7) Since the accommodation case **43** includes the multilayer indication light **62** that indicates the status of the forklift truck **10**, the multilayer indication light **62** does not have to be provided in the vehicle body **11**. As a result, there is no need to process the vehicle body **11** or run wiring to install the multilayer indication light **62**. [0047] (8) Since the automated operation unit **42** is integrated into the box-shaped accommodation case **43** that accommodates most of the device group for the automated operation,

the automated operation unit **42** is not exclusively applicable to the forklift truck **10** of the present embodiment but is applicable to other models. In other words, application of the automated operation unit **42** is not dependent on the model of the forklift truck, and the automated operation unit **42** may be installed on various types of the forklift trucks. Furthermore, by using the automated operation unit **42**, the manufacturing costs of the forklift truck **10** can be significantly reduced as compared to a case where a dedicated counterweight is provided to accommodate most of the device group for the automated operation.

Second Embodiment

[0048] The following will describe a forklift truck according to a second embodiment. In this embodiment, the configuration and position of the automated operation unit are different from those of the first embodiment. In this embodiment, the same configurations as those in the first embodiment will be referred to the description of the first embodiment, and the same reference numerals will be used.

[0049] As illustrated in FIG. 7, a forklift truck **80** has an automated operation unit **81** disposed below the driver's seat **17** in the vehicle body **11**. Specifically, the automated operation unit **81** is installed so as to fit into the recess **41** formed in the front part of the seat stand **39**. The automated operation unit **81** has an accommodation case **82** and the device group for the automated operation that is integrated in the accommodation case **82**.

[0050] The accommodation case **82** is formed by a plurality of plate members into a shape that fits the shape of the recess **41**. The device group for the automated operation integrated into the accommodation case **82** includes the automated operation controller **55**, the touch panel **56**, the changeover switch device **57**, the PLC unit **59**, the DC-DC converter **60**, the wireless communication device **61**, the emergency stop receiver **63**, and the hub **64**. An emergency stop device **83** is provided in the counterweight **16**. In addition, a multilayer indication light **84** is provided on the right rear pillar **33**.

[0051] The forklift truck **80** of the second embodiment offers the same advantages as (1) and (3) of the first embodiment. Furthermore, according to the second embodiment, since the automated operation unit **81** is installed in the recess **41** formed in the front part of the seat stand **39**, the automated operation unit **81** does not have to be displaced relative to the vehicle body **11**. As a result, a displacement mechanism is not necessary, and the manufacturing cost may be reduced.

[0052] The present disclosure is not limited to the above-described embodiments, but may be modified in various manners within the scope of the disclosure, as exemplified below.

[0053] In the above first embodiment, the device group for the automated operation integrated into the accommodation case include the automated operation controller, the touch panel device, the changeover switch device, the emergency stop device, the PLC unit, the DC-DC converter, the wireless communication device, the multilayer indication light, the emergency stop receiver, and the hub, but the configuration is not limited to this. The device group for the automated operation integrated into the accommodation case only have to include at least an automated operation controller, a display device, an input device, and an emergency stop device.

[0054] In the above first embodiment, the automated operation unit is displaced relative to the vehicle body by rotating about the rotation axis as a fulcrum, but the configuration is not limited to this. The automated operation unit may be displaced relative to the vehicle body, for example, by sliding in the front-rear direction, the up-down direction, or the like, relative to the vehicle body. In this case, a guide rail for sliding the automated operation unit only has to be provided on the vehicle body.

[0055] In the above first embodiment, the automated operation unit is installed on the counterweight so as to be located inside the outer edge of the counterweight in a plan view, but the configuration is not limited to this. For example, the accommodation case may be located inside an obstacle detection sensor provided on the counterweight in a plan view. In this case, the automated operation unit may extend outside the counterweight. As a result, a space in the accommodation

case can be increased, and interference between the automated operation unit and obstacles around the vehicle body while the forklift truck travels may be reduced.

[0056] In the above embodiments, the forklift truck is used as an example of the industrial vehicle, but the present disclosure is not limited to the forklift truck. The industrial vehicle only has to be any industrial vehicle capable of manned or unmanned travel other than a forklift truck, and may be, for example, a towing vehicle.

Claims

1. An industrial vehicle capable of manned travel and unmanned travel by automated operation, the industrial vehicle comprising: a vehicle body; and an automated operation unit that is mounted on the vehicle body and in which a plurality of devices for the automated operation is integrated, wherein the plurality of devices includes: an automated operation controller that controls the automated operation; a display device on which information related to the automated operation is displayed; an input device on which data related to the automated operation is input; and an emergency stop device that causes an emergency stop of the automated operation, the automated operation unit includes an accommodation case that accommodates the automated operation controller, the display device, the input device, and the emergency stop device, and the accommodation case is positioned in the vehicle body at a height equal to or lower than a height that does not obstruct an operator's view during the manned travel.
 2. The industrial vehicle according to claim 1, wherein the vehicle body includes: a main body; a driver's seat mounted on the main body; and a counterweight in a rear of the main body, and the accommodation case is mounted on the counterweight and behind the driver's seat.
 3. The industrial vehicle according to claim 2, wherein the accommodation case is positioned at a height equal to or lower than the height of the driver's seat.
 4. The industrial vehicle according to claim 2, wherein the accommodation case is positioned inside an outer edge of the counterweight in a plan view.
 5. The industrial vehicle according to claim 2, wherein the counterweight has an obstacle detection sensor facing rearward, and the accommodation case is disposed inside the obstacle detection sensor in a plan view.
 6. The industrial vehicle according to claim 1, further comprising a displacement mechanism that allows the accommodation case to be displaced relative to the vehicle body.
 7. The industrial vehicle according to claim 1, wherein the emergency stop device includes an emergency stop button protruding from a surface of the accommodation case.
 8. The industrial vehicle according to claim 1, wherein the accommodation case includes a multilayer indication light that indicates a state of the industrial vehicle.
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